Example 1: A Tank Source (Puddle and Pool Fire)

In a small industrial park outside Baton Rouge, Louisiana, a 500-gallon, 4-foot-diameter, vertical tank contains liquid benzene. On June 20, 2016, at 10:30 p.m. local time, a security guard discovers that liquid is leaking out of the tank through a 6-inch circular hole located 10 inches above the bottom of the tank. He also sees that the liquid is flowing onto a paved area in the industrial park. The guard thinks that the tank has just been filled that evening.

The temperature on scene is 80°F, with the wind from the southwest at 7 miles per hour (as measured at a height of 10 meters by a fixed meteorological tower at the site). The sky is more than half covered by clouds and the humidity is about 75 percent. A thunderstorm is approaching from the southwest. There is no low-level inversion. There are very few buildings in the industrial park and a large grassy field is located to the northeast of the industrial park.

The Local Emergency Planning Committee has requested that on-scene responders use ERPG-2 concentrations to define the toxic endpoints in their analysis of benzene hazards.

In this example scenario, you'll determine:

1. Distance to the ERPG-2 level if the puddle evaporates and forms a toxic vapor cloud.
2. Thermal radiation threat if a lightning strike ignites the puddle and forms a pool fire.

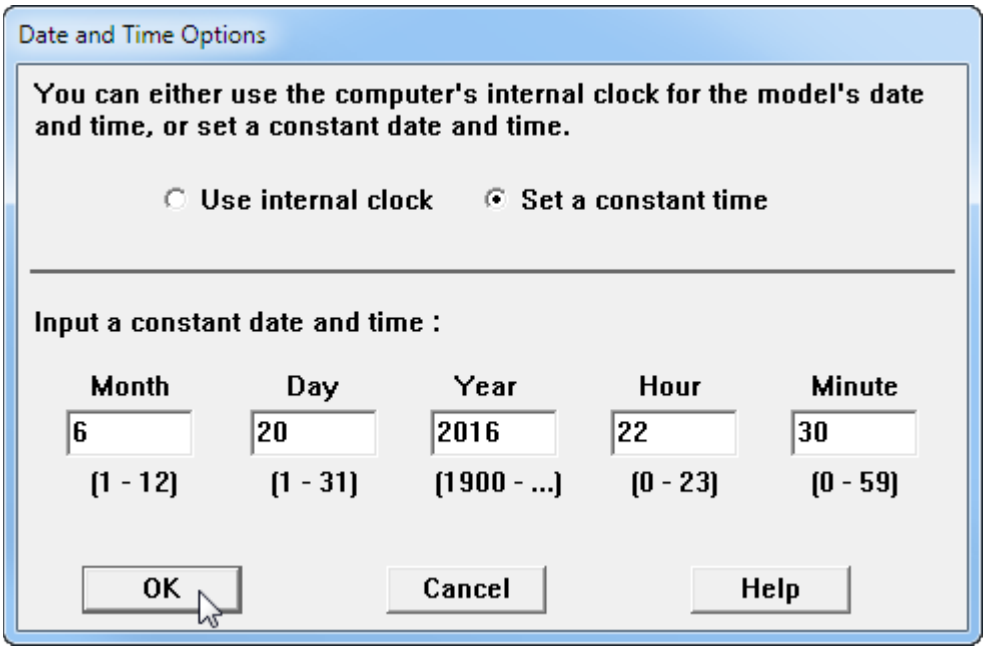
Example 1, Part 1: Modeling a Toxic Vapor Cloud

Choosing a Location and a Chemical

1. Start ALOHA.
2. Read the list of ALOHA's limitations (click Help for more details), then click **OK**.
3. Select **Location** from the **SiteData** menu. A Location Information dialog box appears with a list of the names of cities included in ALOHA's location library.
4. The industrial park is located outside Baton Rouge, Louisiana. Type the characters **ba** to quickly move to the section of the list containing names beginning with "ba." Scroll down a little farther until you see **BATON ROUGE, LOUISIANA**. Click on this name to highlight it, then click **Select**.

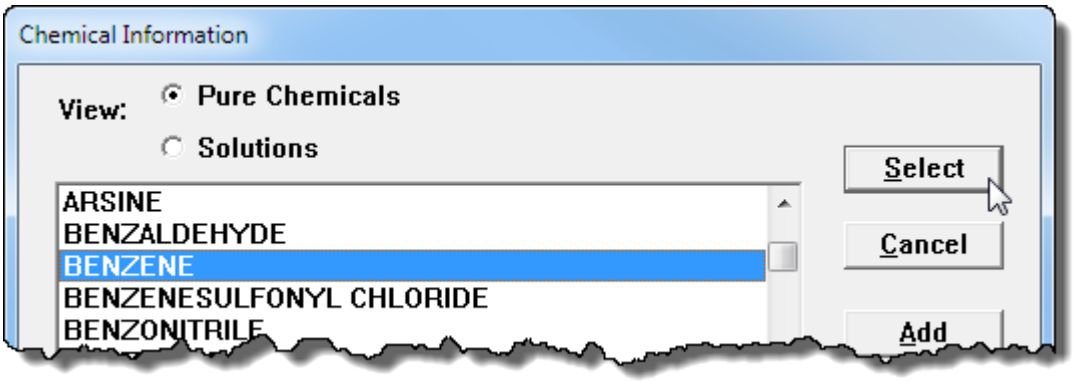
5. Select **Date & Time** from the **SiteData** menu. A Date and Time Options dialog box appears.

6. The release occurs at 10:30 p.m. on June 20, 2016. Select the **Set a constant time** option. Enter the month, day, year, hour, and minute for this scenario. ALOHA requires you to convert the time of day into 24-hour time (click Help to learn how to convert time values). Click **OK**.



7. To choose the chemical that is being released—benzene—select **Chemical** from the **SetUp** menu. A Chemical Information dialog box appears with a list of the chemicals in ALOHA's chemical library.

8. Select **Pure Chemicals** at the top of the window (this should be the default). Find **BENZENE** in the list (type the character **b** to locate benzene more rapidly in the list), click on this name, then click **Select**.



Entering Weather Information and Ground Roughness

Now that you've selected the location, time, and chemical, you must provide information about weather conditions and ground roughness.

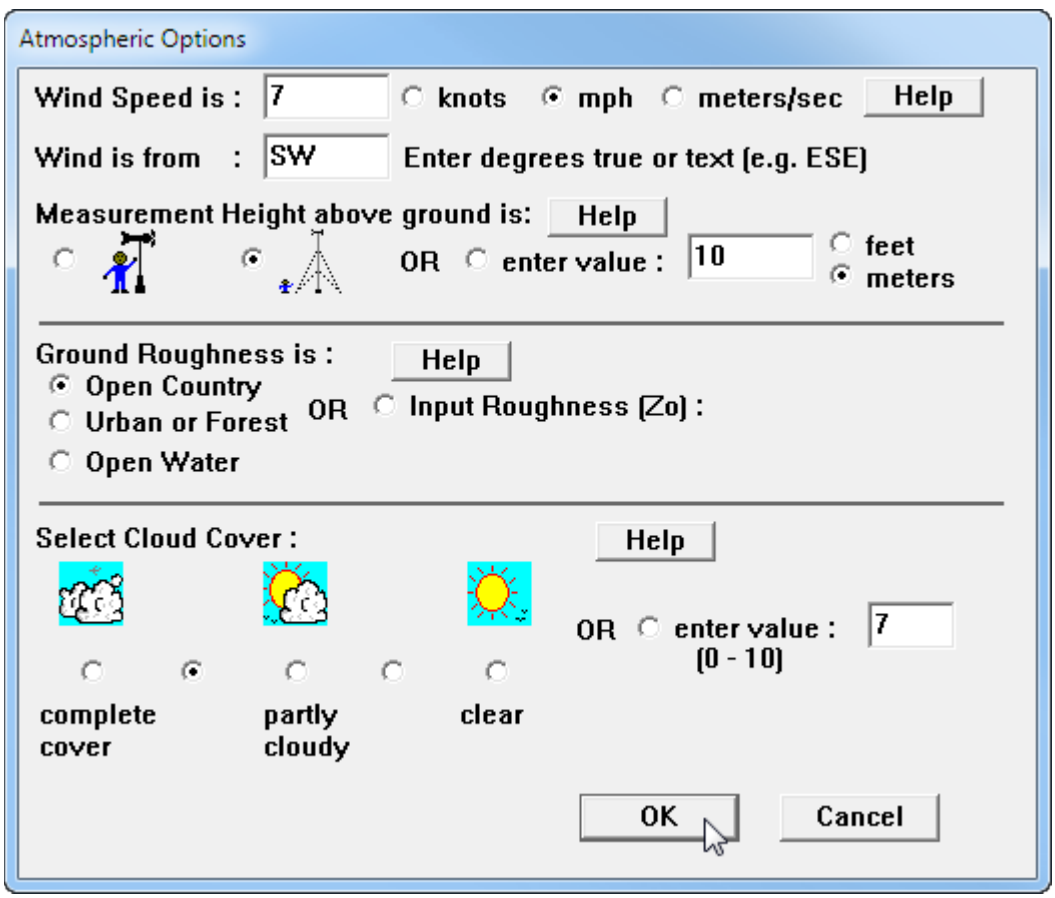
1. In the **SetUp** menu, point to **Atmospheric**, then select **User Input**. The first Atmospheric Options dialog box appears.

2. The wind is travelling from the southwest at a speed of 7 miles per hour. Type **7** in the wind speed box, then select **mph**. Type **SW** in the wind direction box.

3. The wind conditions are measured at a height of 10 meters. Select the **tower icon** in the Measurement Height section. Notice that ALOHA has filled in a value of 10 meters.

4. There are very few buildings in the industrial park and a large grassy field is located to the northeast (the area where the wind would blow the toxic vapor cloud). Select the **Open Country** ground roughness option.

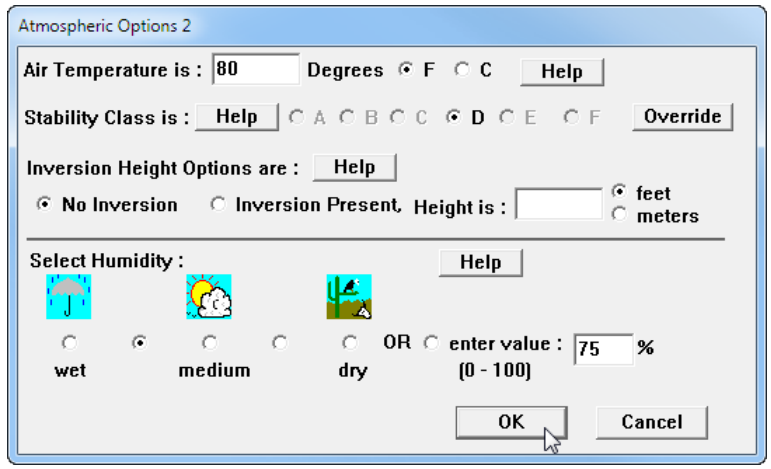
5. The sky is more than half covered by clouds. Under Select Cloud Cover, choose the second option from the left (the option between the complete cover and partly cloudy options). Notice that ALOHA has filled in a value of 7. Click **OK**. The second Atmospheric Options dialog box appears.



6. The air temperature is 80°F. Type **80** in the air temperature box, then select **F**.

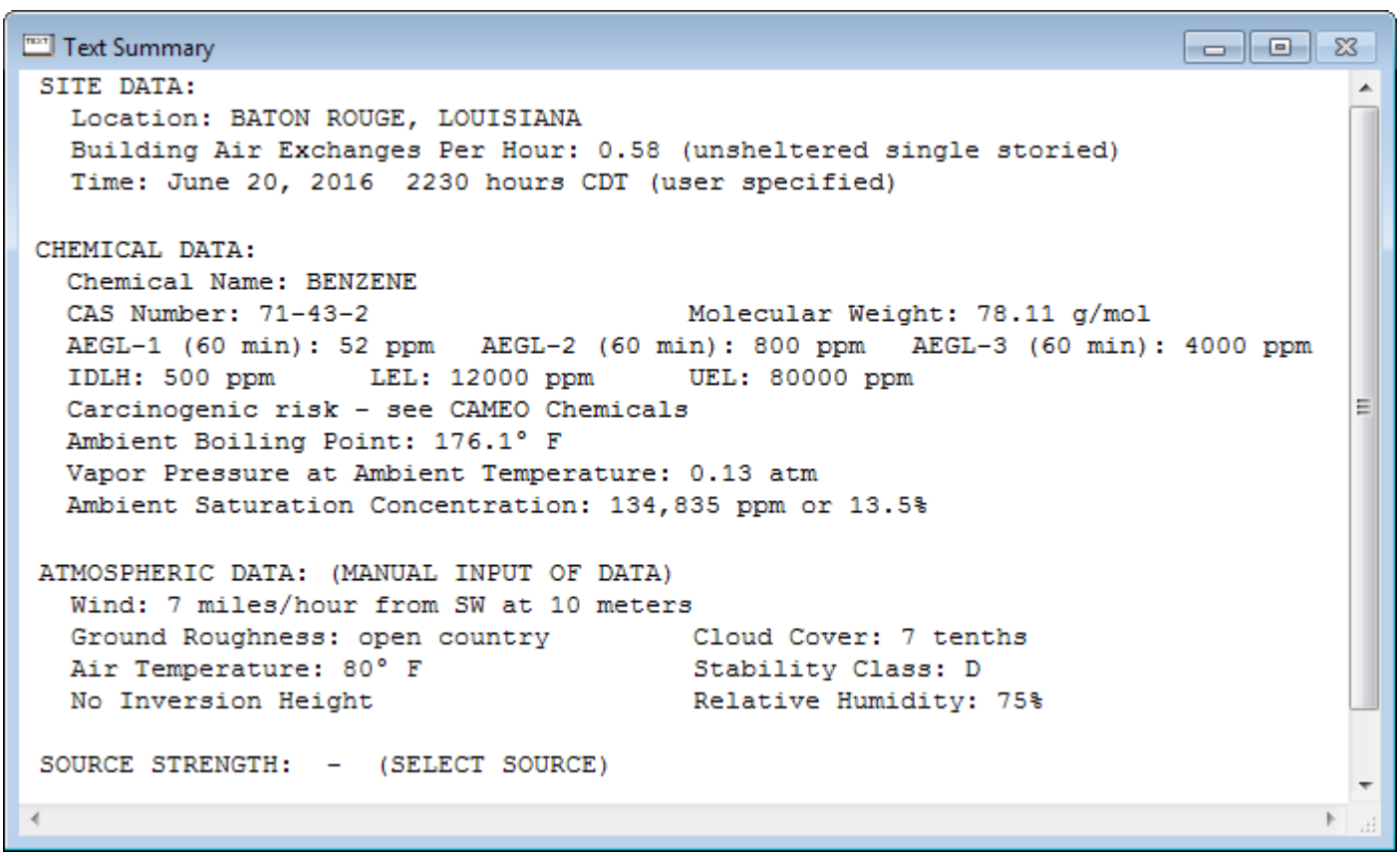
7. ALOHA uses the wind speed, cloud cover, and date and time information that you've entered to automatically select atmospheric **Stability Class D**, representing conditions of neutral atmospheric stability. (Click Help for more details about stability classes and atmospheric stability.)

8. There is no low-level inversion. Check to be sure that **No Inversion** is selected.



9. The relative humidity is about 75 percent. Choose the second option from the left (the option between the wet and medium options). Notice that ALOHA has filled in a value of 75 percent. Click **OK**.

The information that you have entered into ALOHA appears in the Text Summary. Ignore ALOHA's estimate of building exchange rate, since you're not considering infiltration into buildings.



Text Summary

```
SITE DATA:
  Location: BATON ROUGE, LOUISIANA
  Building Air Exchanges Per Hour: 0.58 (unsheltered single storied)
  Time: June 20, 2016  2230 hours CDT (user specified)

CHEMICAL DATA:
  Chemical Name: BENZENE
  CAS Number: 71-43-2                    Molecular Weight: 78.11 g/mol
  AEGL-1 (60 min): 52 ppm   AEGL-2 (60 min): 800 ppm   AEGL-3 (60 min): 4000 ppm
  IDLH: 500 ppm      LEL: 12000 ppm      UEL: 80000 ppm
  Carcinogenic risk - see CAMEO Chemicals
  Ambient Boiling Point: 176.1° F
  Vapor Pressure at Ambient Temperature: 0.13 atm
  Ambient Saturation Concentration: 134,835 ppm or 13.5%

ATMOSPHERIC DATA: (MANUAL INPUT OF DATA)
  Wind: 7 miles/hour from SW at 10 meters
  Ground Roughness: open country         Cloud Cover: 7 tenths
  Air Temperature: 80° F                 Stability Class: D
  No Inversion Height                    Relative Humidity: 75%

SOURCE STRENGTH:  -  (SELECT SOURCE)
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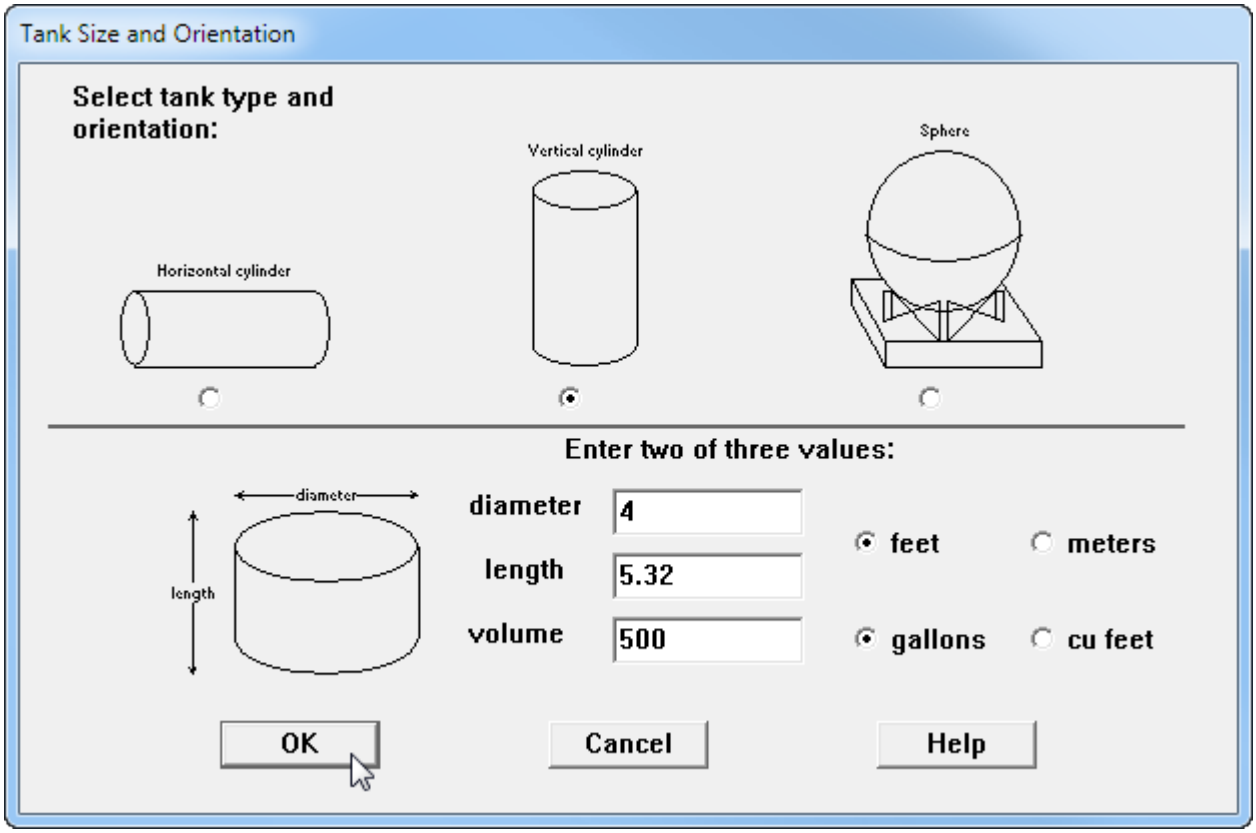
Describing the Release

Now you're ready to enter information about the release itself—that is, to "set the source"—for this release.

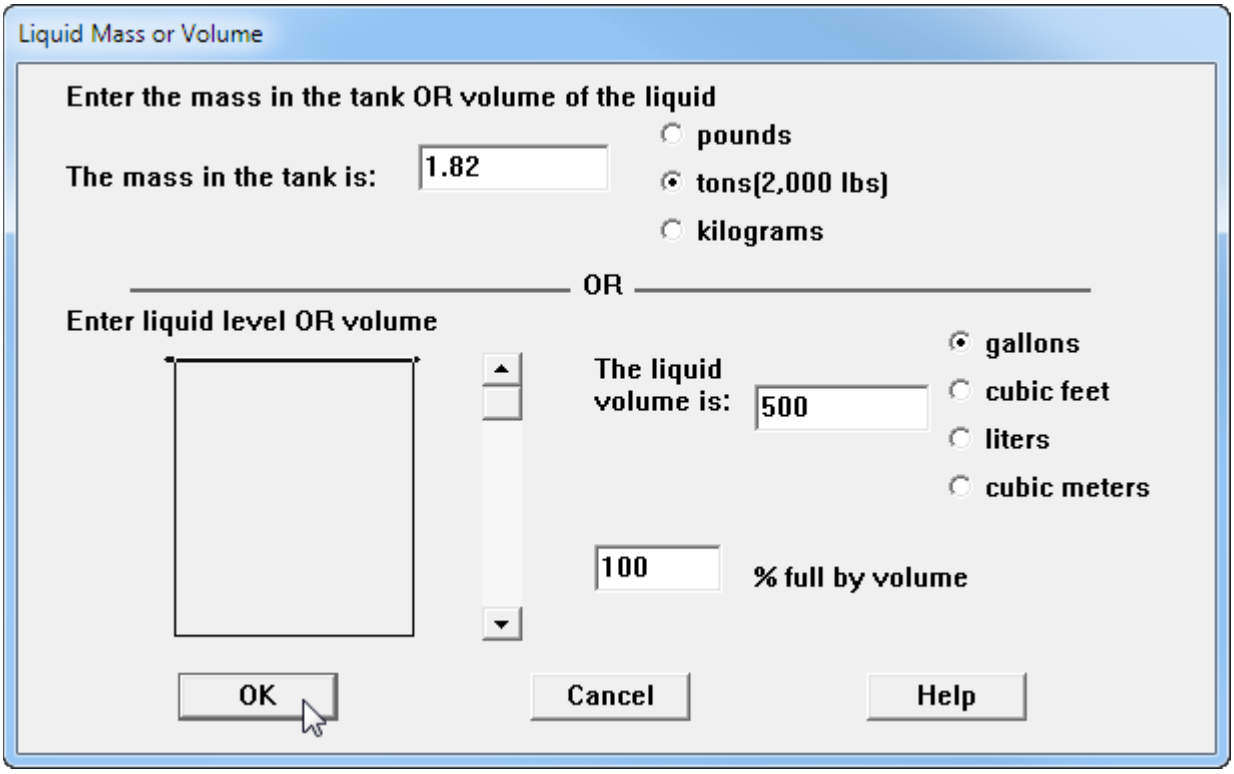
1. The benzene is leaking from a tank. In the **SetUp** menu, point to **Source**, then select **Tank**. A Tank Size and Orientation dialog box appears.

2. The benzene is stored in a 500-gallon, 4-foot-diameter, vertical tank. Select **Vertical cylinder**. Type **500** in the volume box, then select **gallons**. Type **4** in the diameter box, then select **feet**. Notice that ALOHA automatically calculates the tank length. Click **OK**. A Chemical State and Temperature dialog box appears.

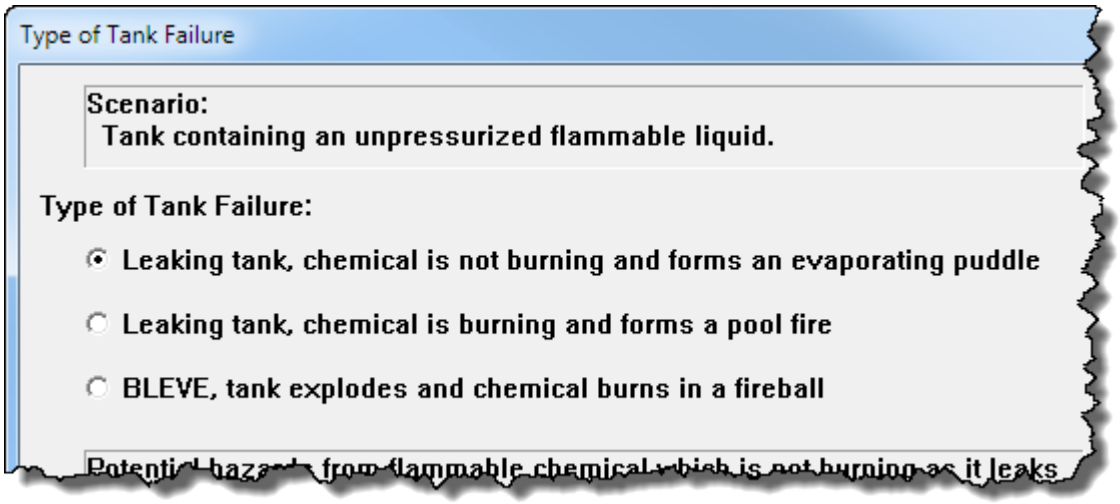


3. The benzene is stored in the tank as a liquid (notice in the Text Summary that it has a boiling point of 176.1°F, which is well above the ambient temperature). Select the **Tank contains liquid** option. Check to be sure that **Chemical stored at ambient temperature** is selected. Click **OK**. A Liquid Mass or Volume dialog box appears.

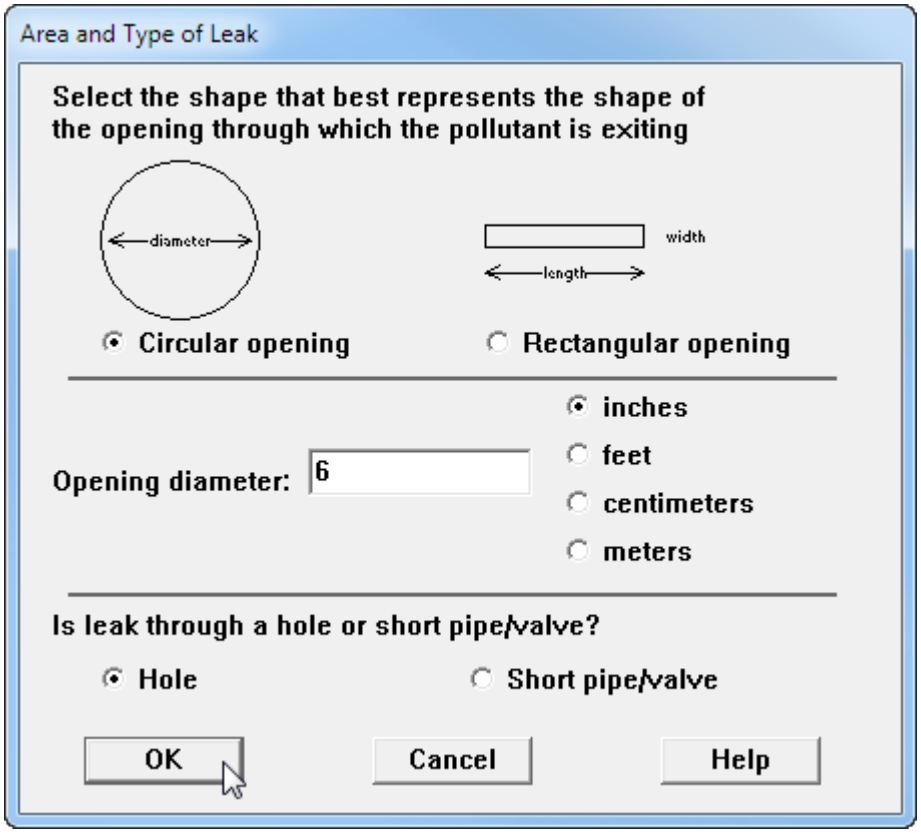
4. The security guard thinks the 500-gallon tank was filled that evening, so the most conservative estimate you can make is that the tank is 100 percent full. Either (a) type **100** in the % full by volume box, (b) type **500** in the liquid volume box, then click **gallons**, or (c) scroll the liquid level bar to the top of the tank diagram. Notice that ALOHA fills in the other values. Click **OK**. A Type of Tank Failure dialog box appears.



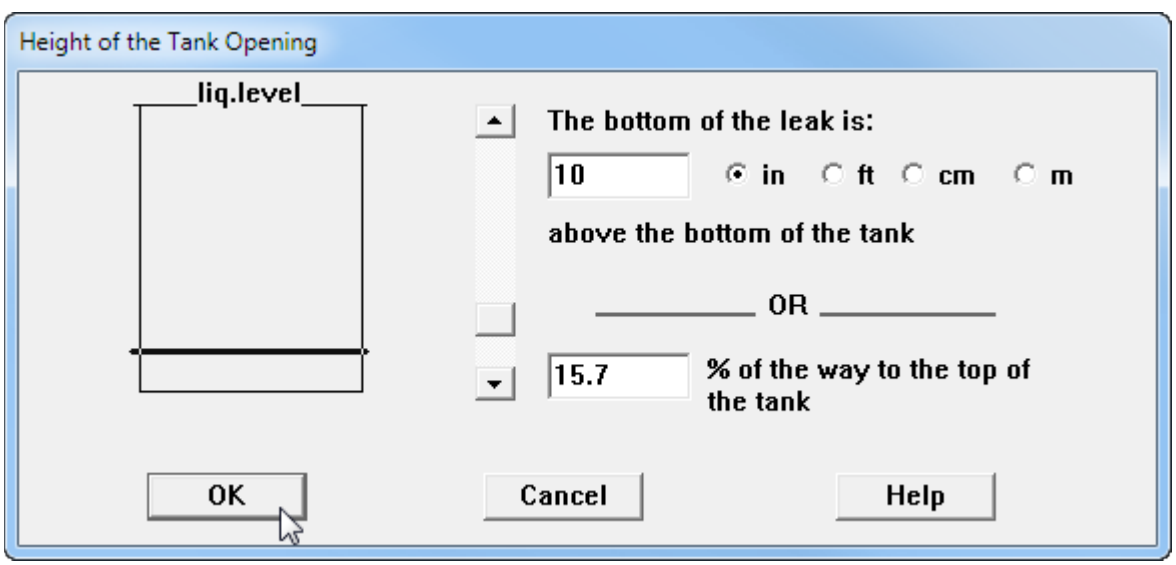
5. Initially, the benzene is leaking from a hole in the tank, but it is not burning. Choose the **Leaking tank, chemical is not burning and forms an evaporating puddle** option. Click **OK**. An Area and Type of Leak dialog box appears.



6. The benzene is leaking from a 6-inch circular hole. Check to be sure that **Circular opening** is selected. Type **6** in the opening diameter box and select **inches**. Choose the **Hole** option. Click **OK**. A Height of the Tank Opening dialog box appears.



7. The hole is 10 inches above the bottom of the tank. Type **10** in the bottom of the leak box and select **in**. Notice that ALOHA fills in the other values. Click **OK**. A Puddle Parameters dialog box appears.



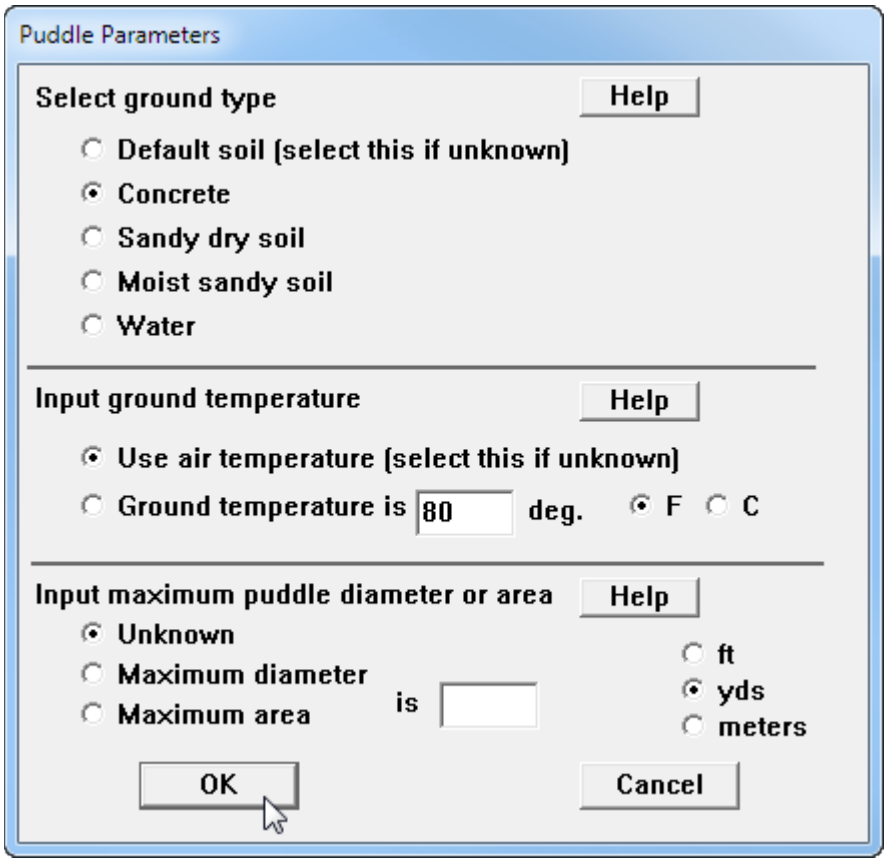
8. The liquid benzene is flowing onto a paved area in the industrial park.

   a. Select the **Concrete** ground type.

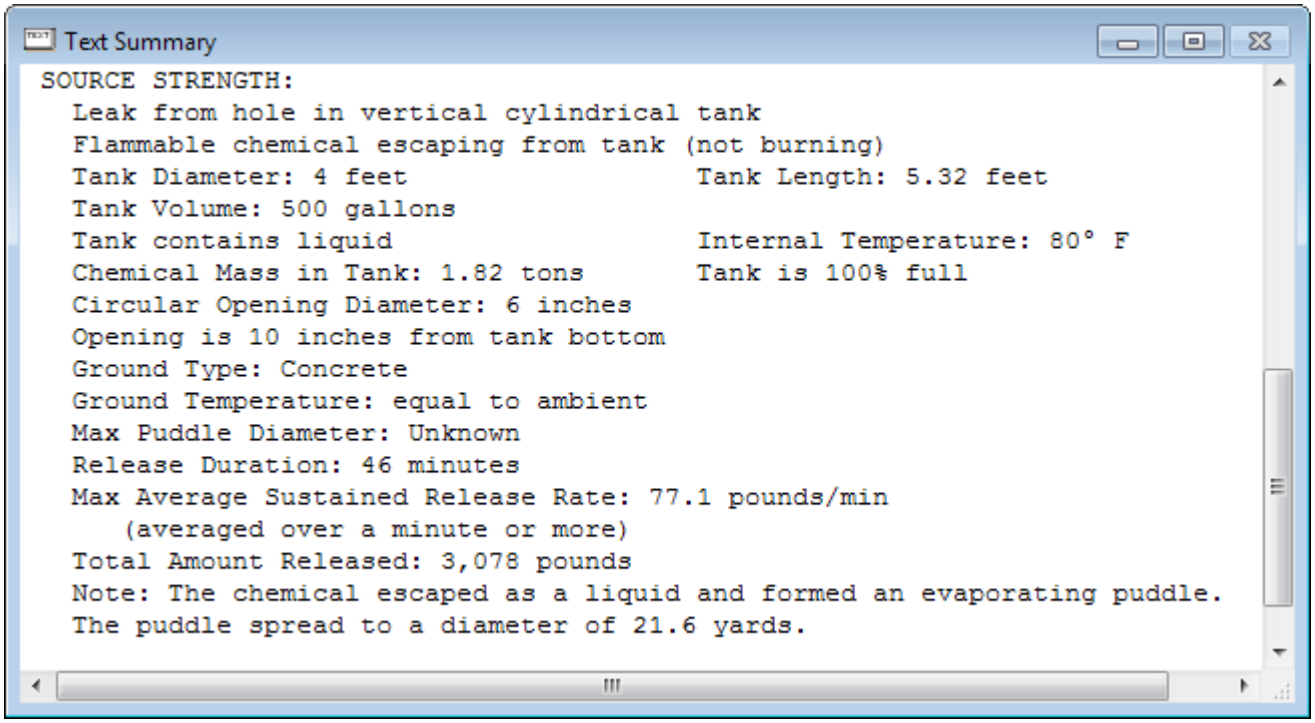
   b. Since you have no information about the ground temperature, select **Use air temperature (select this if unknown)**.

   c. Because the product is flowing onto a paved area, it is probably not contained by a dike, so it will continue spreading outward until it reaches a minimum thickness. Under the "Input maximum puddle diameter or area" heading, click **Unknown**. ALOHA will calculate the area for you based on the release information you provided (up to a maximum diameter of 200 meters for spills on land).

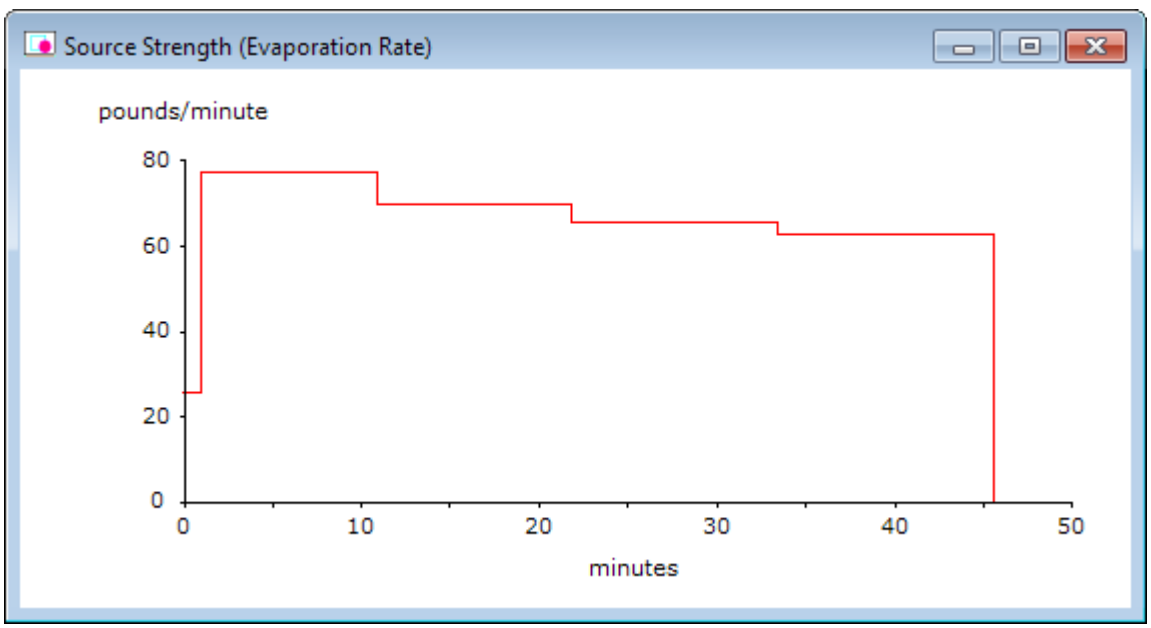
   d. Click **OK**.



The source strength information that you have entered—and the results of ALOHA's source strength calculations—appear in the Text Summary. ALOHA estimates that the release of vapor into the atmosphere lasts for about 46 minutes, and that the maximum amount of vapor released at any one time is 77.1 pounds per minute (this is the Maximum Average Sustained Release Rate). ALOHA estimates that the puddle reached a maximum diameter of 21.6 yards.



9. Choose **Source Strength** from the **Display** menu to see the source strength graph for this scenario. The graph shows the predicted averaged release rate during the hour after the release begins.



Whenever you run ALOHA, ask yourself: Is ALOHA accurately representing what is actually occurring in this scenario? In this case, liquid benzene leaks from a tank to form a puddle; ALOHA expects that because the puddle is undiked, it spreads out to cover a large area and evaporates at a high rate for a relatively short period of time. What if the puddle were constrained by small depressions in the ground? The puddle would not spread out as far because the liquid flowing away from the tank would fill up the

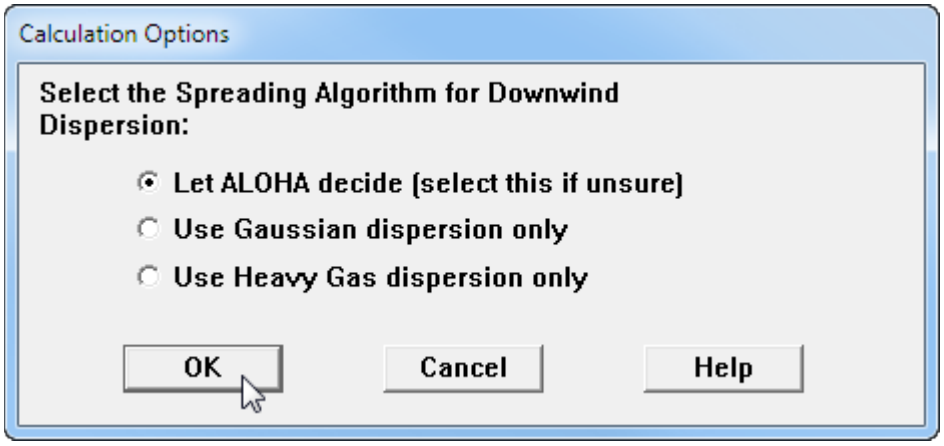
depressions in the ground. The puddle would then be smaller in area and deeper. It would evaporate at a slower rate and it would take longer to completely evaporate.

Because ALOHA assumes that the puddle is on a perfectly flat surface and would spread out until it was very thin, ALOHA may overestimate the real puddle size and evaporation rate. At a real accident scene, check for terrain features that would constrain the puddle from spreading; use this information to estimate the maximum puddle area.

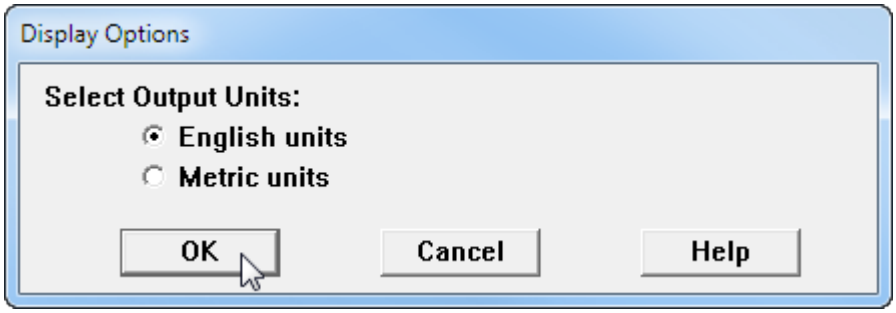
Checking the Model Settings

You don't know if the toxic gas is a heavy gas or not, so you'll want ALOHA to use information about the properties of the chemical and the amount of chemical released to choose whether to make Gaussian or heavy gas dispersion computations. Check to be sure that ALOHA is set to this default.

1. Select **Calculation Options** from the **SetUp** menu. A Calculation Options dialog box appears. Check to be sure that **Let ALOHA decide (select this if unsure)** is selected. Click **OK**.

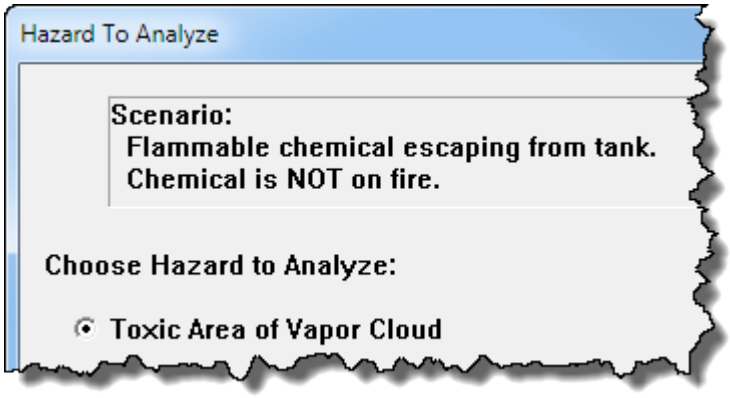


2. Select **Display Options** from the **Display** menu. A Display Options dialog box appears. Select **English units** and ALOHA's computation results will be displayed in those units. Click **OK**.



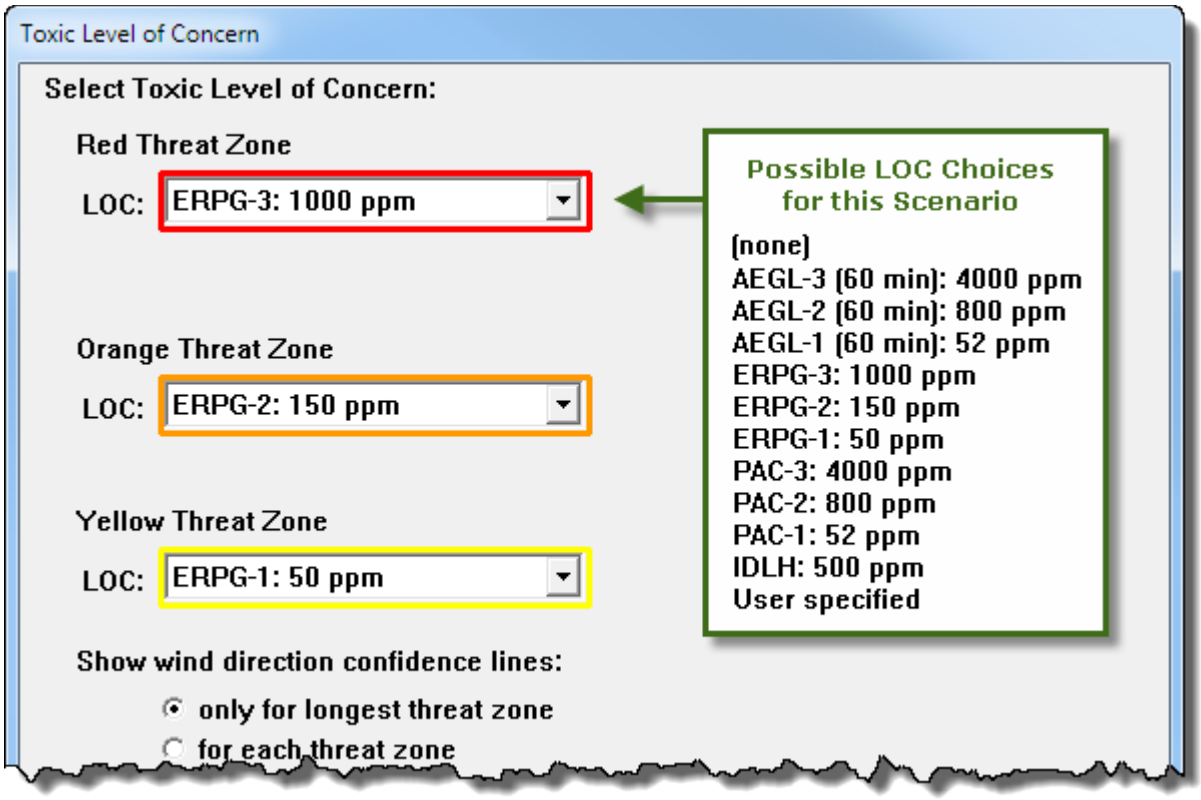
Choosing LOCs and Creating a Threat Zone Estimate

1. Choose **Threat Zone** from the **Display** menu. A Hazard To Analyze dialog box appears.

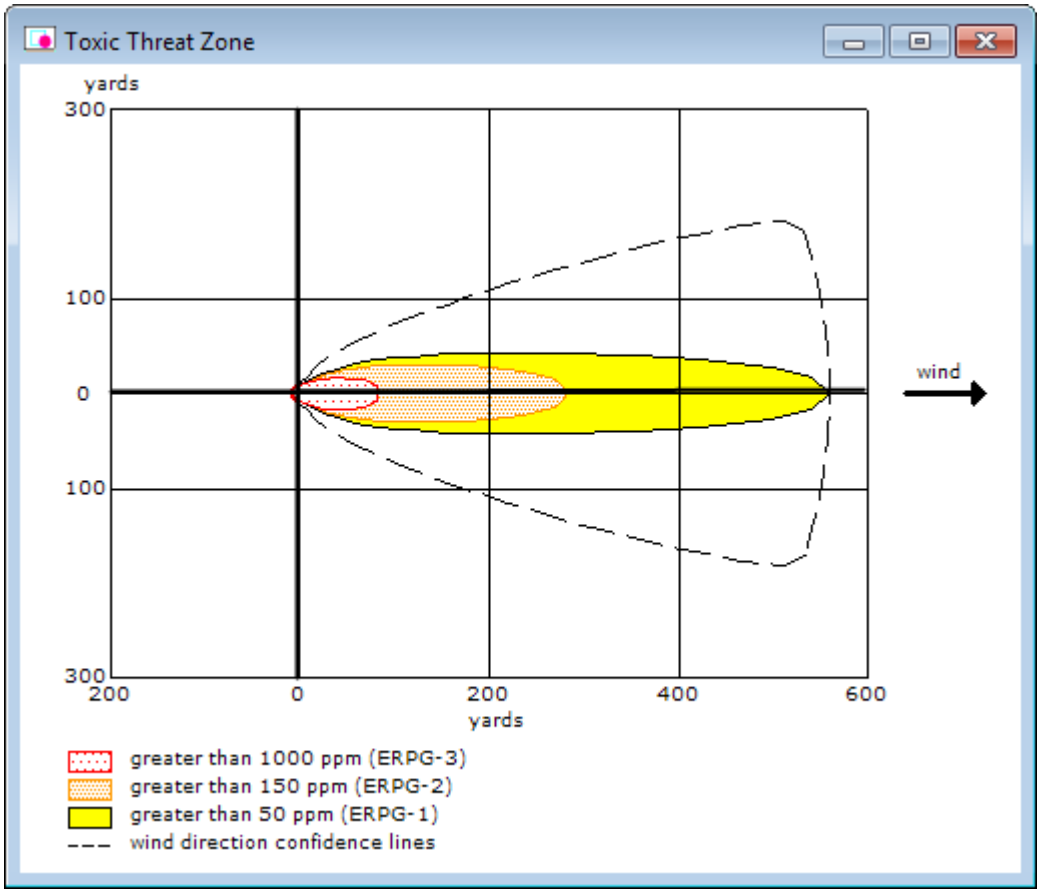


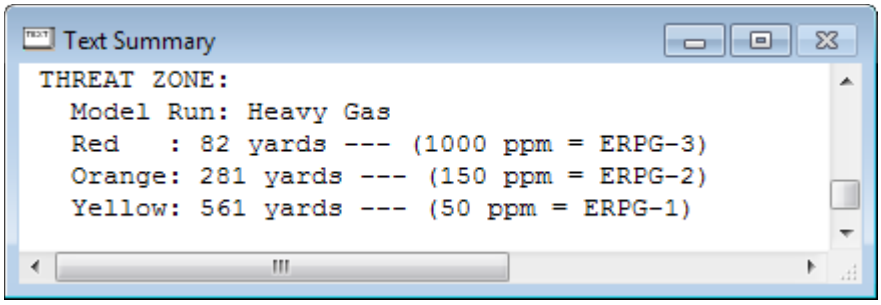
2. As the puddle evaporates, a vapor cloud forms. ALOHA can help you model three possible hazardous scenarios for the flammable vapor cloud: toxic area, flammable area (the area where a flash fire could occur if the cloud encountered an ignition source), or blast area (if a vapor cloud explosion occurred). For this example, you want to display the toxic area threat zone estimate. Select the **Toxic Area of Vapor Cloud** option. Click **OK**. A Toxic Level of Concern dialog box appears.

3. ALOHA uses AEGLs (Acute Exposure Guideline Levels) as the default LOCs for benzene; however, the Local Emergency Planning Committee has requested that on-scene responders use ERPGs (Emergency Response Planning Guidelines). Use the Red Threat Zone LOC drop-down list to change the AEGL-3 value to the ERPG-3 value. Repeat to change the Orange and Yellow Threat Zone LOCs to ERPG-2 and ERPG-1 values, respectively. Check that **Show wind direction confidence lines only for the longest threat zone** has been selected. Click **OK**. ALOHA will display a threat zone estimate for this release.



You'll see ALOHA's threat zone estimate for this scenario. You want to know the downwind distance to the ERPG-2 level specified by the Local Emergency Planning Committee. ALOHA estimates that the orange threat zone will extend 281 yards downwind (the exact value for this *threat distance* is displayed in the Text Summary). Within this zone, ground-level benzene concentrations might exceed the ERPG-2 level. At concentrations above the ERPG-2 level, people could experience serious health effects or find their ability to escape to be impaired (if they are exposed for about an hour).



Note that threat zones displayed in ALOHA are always drawn with the downwind direction to the right—that is, moving across the threat zone window from left to right is the same direction that the wind is moving. It is not an indication of the cardinal wind directions; so, for example, going to the right is not necessarily mean that the wind is coming from the "west". To see the threat zone drawn based on cardinal wind direction, display it in a mapping program (such as MARPLOT).

Check the Text Summary for this release.